STAT 709: Homework 5

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Fall 2026

All exercises are required unless marked optional.

1. Empirical distributions and moment estimates

Bickel and Doksum, *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition, Section 2.5, Problems for Section 2.1: Exercises 6 and 9 (required).

2. Maximum likelihood estimates

Bickel and Doksum, the same edition, Section 2.5, Problems for Section 2.2: Exercises 12 and 13 (required).

3. Method of moments versus maximum likelihood

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Unif}([\theta_1, \theta_2])$, where $n \geq 2$ and both endpoints $\theta_1 < \theta_2$ are unknown. Write $\overline{X} = n^{-1}\sum_{i=1}^n X_i$ and $S^2 = (n-1)^{-1}\sum_i (X_i - \overline{X})^2$. A method-of-moments estimator is

$$\widehat{\theta}_1 = \overline{X} - \sqrt{\frac{3(n-1)}{n} S^2}, \qquad \widehat{\theta}_2 = \overline{X} + \sqrt{\frac{3(n-1)}{n} S^2}.$$

(i) Find the maximum likelihood estimators of θ_1 and θ_2.

(ii) (Optional) Compare the mean squared errors (MSEs) of the method-of-moments estimator and the maximum likelihood estimator. Which estimator is better?